

Lab 7

Bivariate: Correlations and Chi-Square

Walkthrough (5 points)

Instructions

We'll be using the `superman_data.sav` dataset we created last week, to practice with these two statistical models. So, open the dataset and make a quick review of the variables.

RH1: Correlation Walkthrough

Test the research hypothesis that a larger height difference between the actors playing Clark Kent/ Superman and Lois Lane (`height_diff`) is associated with a movie receiving more 'likes' from Letterboxd users (`ldb_likes_thou`, measured in thousands of likes).

The `superman_data.sav` dataset includes the following variables you will use:

- **Height difference (`height_diff`)** – continuous
- **Letterboxd likes in thousands (`ldb_likes_thou`)** – continuous

Data Analysis:

1. **Considering the RH:** Does this research hypothesis suggest a positive, a negative, or no relationship between these variables? Explain.

Answer:

2. **Open the dataset (`superman_data.sav`):** Make sure that...

- Each variable has a label
- All qualitative variables have value labels
- All variables have missing values declared as `-99`

3. **Data Transformation:** The hypothesis is interested in examining the number of likes in thousands. You will want to create a new variable `ldb_likes_thou` that reflects: (`ldb_likes` / 1000).

4. **Data Selection:** The researcher does not want to use a subset of the data.

5. **Examining the Scatterplot:** Follow the handout to get the scatterplot of these variables (put `height_diff` on the Y axis).

- Is the relationship clearly nonlinear? What would we do (or not do) if there was a nonlinear relationship?
- Does the "direction" of the scatterplot seem to support the RH? Why or why not?

Answer:

6. **Statistical analysis:** Follow the handout to get the Pearson's correlation and descriptive.

RH1 Results**Descriptive Statistics:**

Variable	<i>N</i>	<i>M</i>	<i>SD</i>
height_diff	_____	_____	_____
ldb_likes_thou	_____	_____	_____

Correlations:

	<i>r</i>	<i>df</i>	<i>p</i>
height_diff & ldb_likes_thou	_____	_____	_____

N = _____ complete pairs

7. Write-up with univariate stats:

Write-up:

7b. Compose a Table 1 that holds the correlation table for the analysis.

Table 1

Means, Standard Deviations, and Correlations

Variable	<i>M</i>	<i>SD</i>	<i>n</i>	<i>1</i>
1. Height Difference				—
2. Letterboxd Likes (thousands)				

Note. **p* < .05. ***p* < .01. ****p* < .001.

8. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it below this step. **Your TA will not grade the assignment without this.**

Paste Homework Checker screenshot here:

RH2: Chi-Square Walkthrough

Test the hypothesis that, among live-action Superman actors, those over 6ft tall are more likely to portray the character in media rated as 'Fresh' by critics on Rotten Tomatoes, whereas actors under 6ft tall are equally likely to appear in 'Fresh' or 'Rotten' media.

The superman_data.sav dataset includes the following variables you will use:

- **Actor height group (clark_grp)** - categorical/nominal; 2-levels
- **Tomatometer rating (tomatometer)** - categorical/nominal; 2-levels

Data Analysis:**1. Considering the RH:**

- a. Label the contingency table to depict the RH by changing each ? to <, > or =.

Table 2

Actor height	Fresh	Use: >, <, =	Rotten
6ft or taller		?	
Under 6ft		?	

- b. Does this research hypothesis suggest a symmetrical pattern of relationship, asymmetrical pattern of relationship or no relationship between these variables?

Answer:

2. **Examining the Contingency Table:** Follow the handout to get the cross-tabulation & fill in the above.

- Does the “pattern” in the table seem to support the RH? Why or why not?

Answer:

3. **Statistical analysis:** Follow the handout to get the Pearson’s X^2 between these variables.

Number of 6ft or taller in the sample: ____

Number of Under 6ft in the sample: ____

Number of Fresh in the sample: ____

Number of Rotten in the sample: ____

$\chi^2 =$ ____ $df =$ ____ $p =$ ____

State the H_0 : There is no pattern of relationship between clark_grp and tomatometer

Retain or reject H_0 ? ____

Support research hypothesis? ____

4. Compose a Table 2 that holds the contingency table for the analysis.

Table 3

Actor height	Fresh	Rotten	Total
6ft or taller			
Under 6ft			
Total			

5. **Write-up the analysis:** Follow the example in the SPSS how-to. Refer to Table 2 in your write-up.

Write-up:

6. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it below this step. **Your TA will not grade the assignment without this.**

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Your Turn (20 points)

Instructions

In the next section, you will practice using these statistical models on your own using a new dataset. We'll be using the `hotones_data.sav` dataset, that includes information related to the interview show Hot Ones where celebrity guests are interviewed over a platter of increasingly spicy chicken wings.

The show features 10 progressively spicier chicken wings, each coated with a different hot sauce. The spiciness of each sauce is measured using the **Scoville Heat Unit (SHU)** scale, which measures the concentration of capsaicin (the chemical compound that makes peppers hot). For reference, a jalapeño pepper ranges from 2,500 to 8,000 SHU, while some of the hottest sauces on Hot Ones can exceed 2,000,000 SHU!

Open the dataset and make a quick review of the variables.

RH3: Correlation - Your Turn #1

Test the research hypothesis that, among episodes where guests completed the challenge without quitting, episodes where guests consumed higher total Scoville Heat Units (`total_SHU_mil`, measured in millions) would be associated with more views on YouTube (`views`).

The `hotones_data.sav` dataset includes the following variables you will use:

- **Total SHU in millions** (`total_SHU_mil`) – continuous
- **YouTube views** (`views`) – continuous

Data Analysis:

1. **Considering the RH:** Does this research hypothesis suggest a positive, a negative, or no relationship between these variables? Explain.

Answer:

2. **Open the dataset (`hotones_data.sav`):** Make sure that...

- Each variable has a label
- All qualitative variables have value labels
- All variables have missing values declared as `-99`

3. **Data Transformation:** Compute `total_SHU_mil` by summing all 10 sauce SHU values and dividing by 1,000,000.

4. **Data Selection:** Select only those episodes where `result = 1`.

5. **Examining the Scatterplot:** Follow the handout to get the scatterplot (put `total_SHU_mil` on the Y axis).

- Is the relationship clearly nonlinear?
- Does the direction support the RH?

Answer:

6. **Statistical analysis:** Follow the handout to get the Pearson's correlation and descriptive.

Your Turn #1 Results

Descriptive Statistics:

Variable	<i>N</i>	<i>M</i>	<i>SD</i>
total_SHU_mil	_____	_____	_____
views	_____	_____	_____

Correlations:

	<i>r</i>	<i>df</i>	<i>p</i>
total_SHU_mil & views	_____	_____	_____

N = _____ complete pairs

7. Write-up with univariate stats:

Write-up:

8. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it below this step. **Your TA will not grade the assignment without this.**

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RH4: Correlation - Your Turn #2

Test the research hypothesis that, among episodes where guests completed the challenge without quitting, guests that consumed higher total Scoville Heat Units (*total_SHU_mil*, measured in millions) would be associated with less engagement (comments per view) on the video (*engagement*).

The *hotones_data.sav* dataset includes the following variables you will use:

- **Total SHU in millions (*total_SHU_mil*)** – continuous
- **Engagement - comments per view (*engagement*)** – continuous

Data Analysis:

1. **Considering the RH:** Does this research hypothesis suggest a positive, a negative, or no relationship between these variables? Explain.

Answer:

2. **Open the dataset (*hotones_data.sav*):** Make sure that...

- Each variable has a label
- All qualitative variables have value labels
- All variables have missing values declared as -99

3. **Data Transformation:** Create $\text{engagement} = \text{comments} / \text{views}$.

4. **Data Selection:** Select only those episodes where `result = 1`.

5. **Examining the Scatterplot:** Follow the handout to get the scatterplot (put `total_SHU_mil` on the Y axis).

Answer:

6. **Statistical analysis:** Follow the handout to get the Pearson's correlation and descriptive.

Your Turn #2 Results

Descriptive Statistics:

Variable	<i>N</i>	<i>M</i>	<i>SD</i>
total_SHU_mil	_____	_____	_____
engagement	_____	_____	_____

Correlations:

	<i>r</i>	<i>df</i>	<i>p</i>
total_SHU_mil & engagement	_____	_____	_____

N = _____ complete pairs

7. Write-up with univariate stats:

Write-up:

8. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it below this step. **Your TA will not grade the assignment without this.**

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9. Finally, create an APA-style correlation table (Table 3) that includes the results of your analyses for both RH3 and RH4.

Table 4

Means, Standard Deviations, and Correlations

Variable	<i>M</i>	<i>SD</i>	<i>n</i>	1	2
1. Views				—	—
2. Engagement					—
3. Total SHU (millions)					

Note. **p* < .05. ***p* < .01. ****p* < .001.

RH5: Chi-Square - Your Turn #1

Test the research hypothesis that, among episodes where guests completed the challenge without quitting, actors are more likely to appear on less popular episodes ($\leq 125,000$ likes), whereas non-actors are equally likely to appear on less popular or more popular episodes.

The hotones_data.sav dataset includes the following variables you will use:

- **Guest occupation type (actor)** - categorical/nominal; 2-levels
- **Episode popularity (popular)** - categorical/nominal; 2-levels

Data Analysis:

1. Considering the RH:

- a. Label the contingency table to depict the RH by changing each ? to <, > or =.

Table 5

Occupation Type	Less Popular ($\leq 125k$)	Use: >, <, =	More Popular ($> 125k$)
Non-Actor		?	
Actor		?	

- b. Does this research hypothesis suggest a symmetrical pattern of relationship, asymmetrical pattern of relationship or no relationship between these variables?

Answer:

2. **Data Transformation:** Recode occupation into actor and likes into popular as described above.

3. **Data Selection:** Select only those episodes where result = 1.

4. **Examining the Contingency Table:** Follow the handout to get the cross-tabulation.

- Does the pattern seem to support the RH?

Answer:

5. **Statistical analysis:** Follow the handout to get the Pearson's χ^2 between these variables.

Number of Non-Actor in the sample: ____

Number of Actor in the sample: ____

Number of Less Popular ($\leq 125k$) in the sample: ____

Number of More Popular ($> 125k$) in the sample: ____

$\chi^2 =$ ____ $df =$ ____ $p =$ ____

State the H_0 : There is no pattern of relationship between actor and popular

Retain or reject H_0 ? ____

Support research hypothesis? ____

6. Compose a Table 4 that holds the contingency table for the analysis.

Table 6

Occupation Type	Less Popular ($\leq 125k$)	More Popular ($> 125k$)	Total
Non-Actor			
Actor			
Total			

7. **Write-up the analysis:** Refer to Table 4 in your write-up.

Write-up:

8. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it below this step. **Your TA will not grade the assignment without this.**

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RH6: Chi-Square - Your Turn #2

Test the research hypothesis that, among episodes where guests completed the challenge without quitting, younger guests (under 35) are equally likely to appear on less popular ($\leq 125k$ likes) or more popular episodes ($> 125k$ likes), whereas older guests (35+) are more likely to appear on more popular episodes.

The `hotones_data.sav` dataset includes the following variables you will use:

- **Guest age group (`age_grp`)** - categorical/nominal; 2-levels
- **Episode popularity (`popular`)** - categorical/nominal; 2-levels

Data Analysis:

1. Considering the RH:

- Label the contingency table to depict the RH by changing each ? to $<$, $>$ or $=$.

Table 7

Age Group	Less Popular ($\leq 125k$)	Use: $>$, $<$, $=$	More Popular ($> 125k$)
Under 35		?	
35 and Over		?	

- Does this research hypothesis suggest a symmetrical pattern of relationship, asymmetrical pattern of relationship or no relationship between these variables?

Answer:

2. **Data Transformation:** Create `age_grp` from `age` (under 35 = 1, 35 and over = 2).

3. **Data Selection:** Select only those episodes where `result` = 1.

4. **Examining the Contingency Table:** Follow the handout to get the cross-tabulation.

- Does the pattern seem to support the RH?

Answer:

5. **Statistical analysis:** Follow the handout to get the Pearson's X^2 between these variables.

Number of Under 35 in the sample: ____

Number of 35 and Over in the sample: ____

Number of Less Popular ($\leq 125k$) in the sample: ____

Number of More Popular ($> 125k$) in the sample: ____

$\chi^2 =$ ____ $df =$ ____ $p =$ ____

State the H_0 : There is no pattern of relationship between age_grp and popular

Retain or reject H_0 ? ____

Support research hypothesis? ____

6. Compose a Table 5 that holds the contingency table for the analysis.

Table 8

Age Group	Less Popular ($\leq 125k$)	More Popular ($> 125k$)	Total
Under 35			
35 and Over			
Total			

7. **Write-up the analysis:** Refer to Table 5 in your write-up.

Write-up:

8. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it below this step. **Your TA will not grade the assignment without this.**

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